

Systems Thinking Project

Acrobot

Phase 2: Stability Analysis

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1 Introduction

1.1 What is an Acrobot?

An acrobot is a two-link pendulum that hangs from a fixed pivot. The first link (the "shoulder" link) is free to rotate about the pivot; the second link is joined to the end of the first at the "elbow". The only motor sits at the elbow, so the shoulder is *passive*. It is the robot version of a gymnast on a high bar who can only swing the legs at the hips. Because there are two degrees of freedom but only one actuator, the acrobot is a standard example of an *underactuated* system [1], [2].

1.2 Model

The configuration is described by two generalised coordinates: q_1, the angle of link 1 from the downward vertical, and q_2, the angle of link 2 relative to link 1. The full state contains the angles and their rates:

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}.$$

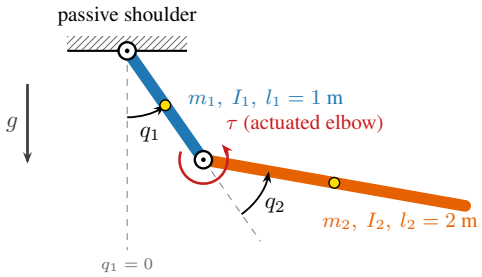


Figure 1: The acrobot: passive shoulder at the pivot, motor torque τ at the elbow; dots mark the centres of mass.

1.3 Equations of Motion

The dynamics (given in the Phase 1 helper and verified in Phase 1) are

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} = \tau_g(q) + B\,\tau, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad |\tau| \le 20\,\mathrm{N\,m}. \tag{1}$$

τ **(Input Torque):** the torque of the elbow motor, the only quantity we command. $B = [0,\ 1]^T$ says it acts on the elbow equation only.

$M(q)$ **(Inertia Matrix):** the resistance to angular acceleration. It depends on the elbow angle because the mass distribution changes as the arm folds:

$$M(q) = \begin{bmatrix} 2.663 + 2\cos q_2 & 1.33 + \cos q_2 \\ 1.33 + \cos q_2 & 1.33 \end{bmatrix}, \qquad \det M = 1.7729 - \cos^2 q_2 \ge 0.7729 > 0. \tag{2}$$

$C(q,\dot{q})$ **(Coriolis Matrix):** the coupling forces between the links (centrifugal and Coriolis effects):

$$C(q,\dot{q}) = \sin q_2 \begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{3}$$

$\tau_g(q)$ **(Gravity Torque):** the torque that gravity exerts on each joint, $\tau_g = -\partial U/\partial q$, with potential energy U:

$$U = -14.715\cos q_1 - 9.81\cos(q_1+q_2), \qquad \tau_g(q) = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin(q_1+q_2) \\ \sin(q_1+q_2) \end{bmatrix}. \tag{4}$$

System Parameters:

$$m_1 = m_2 = 1\,\mathrm{kg}, \qquad l_1 = 1\,\mathrm{m}, \qquad l_2 = 2\,\mathrm{m}, \qquad l_{c1} = 0.5\,\mathrm{m}, \qquad l_{c2} = 1\,\mathrm{m},$$
$$I_1 = 0.083\,\mathrm{kg\,m^2}, \qquad I_2 = 0.33\,\mathrm{kg\,m^2}, \qquad g = 9.81\,\mathrm{m\,s^{-2}}.$$

1.4 Equilibria and Linear Model (from Phase 1)

With $\tau^* = 0$ the equilibria satisfy $\tau_g(q^*) = 0$, i.e. $\sin q_1^* = 0$ and $\sin(q_1^* + q_2^*) = 0$. This gives four postures:

Table 1: The four equilibria

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0, 0)$	-24.525	both links hanging
E_2	down–up	$(0, \pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi, 0)$	$+24.525$	fully upright (target)
E_4	up–down	(π, π)	$+4.905$	link 1 up, link 2 down

Linearizing about each equilibrium with $\tilde{x} = x - x^*$ gives

$$\dot{\tilde{x}} = A\tilde{x} + B_\ell\tilde{\tau}, \qquad A = \begin{bmatrix} 0 & I \\ P & 0 \end{bmatrix}, \qquad B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix}, \tag{5}$$

with $P = M(q^*)^{-1}K(q^*)$ and $r = M(q^*)^{-1}B$, where $K = \partial\tau_g/\partial q$. The transfer functions from τ to the angles are

$$G_1(s) = \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \qquad G_2(s) = \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)}, \tag{6}$$
$$\Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P.$$

Table 2: Transfer functions at the four equilibria (Phase 1)

	$\Delta(s)$	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$
E_1	$s^4 + 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 + 4.2103)$	$6.0332\,(s^2 + 5.2595)$
E_2	$s^4 + 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 - 29.7273)$	$0.8578\,(s^2 + 7.3982)$
E_3	$s^4 - 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 - 4.2103)$	$6.0332\,(s^2 - 5.2595)$
E_4	$s^4 - 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 + 29.7273)$	$0.8578\,(s^2 - 7.3982)$

1.5 Objective

Determine the stability of every equilibrium using both the transfer functions and the state-space model, and support each conclusion with simulation. At the upright equilibrium E_3, determine whether feedback from a *single* measured joint angle can stabilize the system, and explain why.

2 Stability Analysis

2.1 Stability Criteria

- **State space:** the linear model (5) is *asymptotically stable* if every eigenvalue of A has $\operatorname{Re}\lambda < 0$, *unstable* if any has $\operatorname{Re}\lambda > 0$, and *marginally stable* if the eigenvalues on the imaginary axis are simple and none lies to the right.
- **Transfer function:** G_j is *BIBO stable* if every pole has $\operatorname{Re} p < 0$.
- **Nonlinear system (Lyapunov's indirect method):** an eigenvalue with $\operatorname{Re}\lambda > 0$ makes the equilibrium of (1) unstable; eigenvalues on the imaginary axis leave the question open [4].

2.2 Poles of the Transfer Functions = Eigenvalues of A

At every equilibrium

$$\operatorname{rank}[B_\ell \;\; AB_\ell \;\; A^2B_\ell \;\; A^3B_\ell] = 4, \qquad \operatorname{rank}\begin{bmatrix} C_j \\ C_jA \\ C_jA^2 \\ C_jA^3 \end{bmatrix} = 4 \quad (j = 1, 2),$$

so both G_1 and G_2 are minimal realizations with no pole–zero cancellation. Their poles are exactly the eigenvalues of A, and the transfer-function and state-space verdicts must coincide.

2.3 Characteristic Polynomial

$$\Delta(s) = s^4 + 0 \cdot s^3 - \operatorname{tr}(P)\, s^2 + 0 \cdot s + \det P. \tag{7}$$

A Hurwitz polynomial has all coefficients positive. Here the s^3 and s^1 coefficients are zero because the model has no friction, so **no equilibrium is asymptotically stable with** $\tau = 0$. Writing $\lambda = s^2$ (the eigenvalues of P):

$$\lambda^2 - \operatorname{tr}(P)\lambda + \det P = 0, \qquad \lambda < 0 \Rightarrow s = \pm j\sqrt{-\lambda}, \qquad \lambda > 0 \Rightarrow s = \pm\sqrt{\lambda}. \tag{8}$$

- $\det P > 0$, $\operatorname{tr} P < 0$: both $\lambda < 0$, all poles on the $j\omega$-axis (E_1).
- $\det P < 0$: one λ of each sign, one real RHP pole (E_2, E_4).
- $\det P > 0$, $\operatorname{tr} P > 0$: both $\lambda > 0$, two RHP poles (E_3).

2.4 Link to the Potential Energy

Since $K = -\nabla^2 U$ and $M \succ 0$, $P = M^{-1}K$ is similar to $M^{-1/2}KM^{-1/2}$, which is congruent to K. By Sylvester's law of inertia,

$$\#\{\text{RHP poles}\} = \#\{\text{negative eigenvalues of } \nabla^2 U\}. \tag{9}$$

A minimum of U gives no RHP pole, a saddle one, and a maximum two [3].

2.5 Stability of Each Equilibrium

Table 3: Stability of the four equilibria

	$(\operatorname{tr} P, \det P)$	poles of $G_1, G_2 = \operatorname{eig}(A)$	$\operatorname{eig}\nabla^2 U$	RHP	Verdict
E_1	$(-42.24,\ 186.77)$	$\pm j2.2400,\ \pm j6.1011$	4.91, 29.43 (min)	0	marginally stable
E_2	$(-8.40,\ -186.77)$	$\pm 3.1775,\ \pm j4.3010$	-14.72, 9.81 (saddle)	1	unstable
E_3	$(42.24,\ 186.77)$	$\pm 2.2400,\ \pm 6.1011$	-29.43, -4.91 (max)	2	unstable
E_4	$(8.40,\ -186.77)$	$\pm 4.3010,\ \pm j3.1775$	-9.81, 14.72 (saddle)	1	unstable

E_1 (down–down).

- **Transfer function:** the poles are simple and on the imaginary axis, so G_1 and G_2 are *not* BIBO stable. A bounded input at a resonance, $\tau = \sin(2.24t)$, gives a double pole at $\pm j2.24$ and a response that grows like $t\sin(2.24t)$.
- **State space:** the free response is a sum of two undamped modes ($\omega = 2.24$ and 6.10 rad/s): bounded but not decaying, so **marginally stable**.
- **Nonlinear:** linearization is inconclusive, so we use energy as a Lyapunov function:

$$V(x) = \tfrac{1}{2}\dot{q}^T M(q)\dot{q} + U(q) - U(q^*), \qquad \nabla^2 U(q^*) \succ 0 \Rightarrow V \succ 0, \qquad \dot{V} = \dot{q}^T B\tau = 0. \tag{10}$$

  E_1 is **stable in the sense of Lyapunov but not asymptotically stable**: the swing never decays.

E_2 and E_4 (one link up). $\det P = -186.77 < 0$, so each has one real RHP pole ($+3.1775$ at E_2, $+4.3010$ at E_4). Both are BIBO unstable, A is unstable, and by Lyapunov's indirect method the nonlinear equilibria are unstable.

E_3 (up–up). Gravity flips, $P_{E3} = -P_{E1}$, so λ changes sign and the poles become ± 2.2400, ± 6.1011: **two RHP poles**, unstable in every sense.

2.6 Simulation Evidence

All Phase 2–4 simulations use the full nonlinear model in **PyBullet** [5] (uniform rods, no damping, step 0.25 ms). Checked against an RK4 integration of (1) with $q(0) = (0.8, -0.5)$ rad and $\tau = 3\sin 2t$, the two agree to within 0.52° over 5 s.

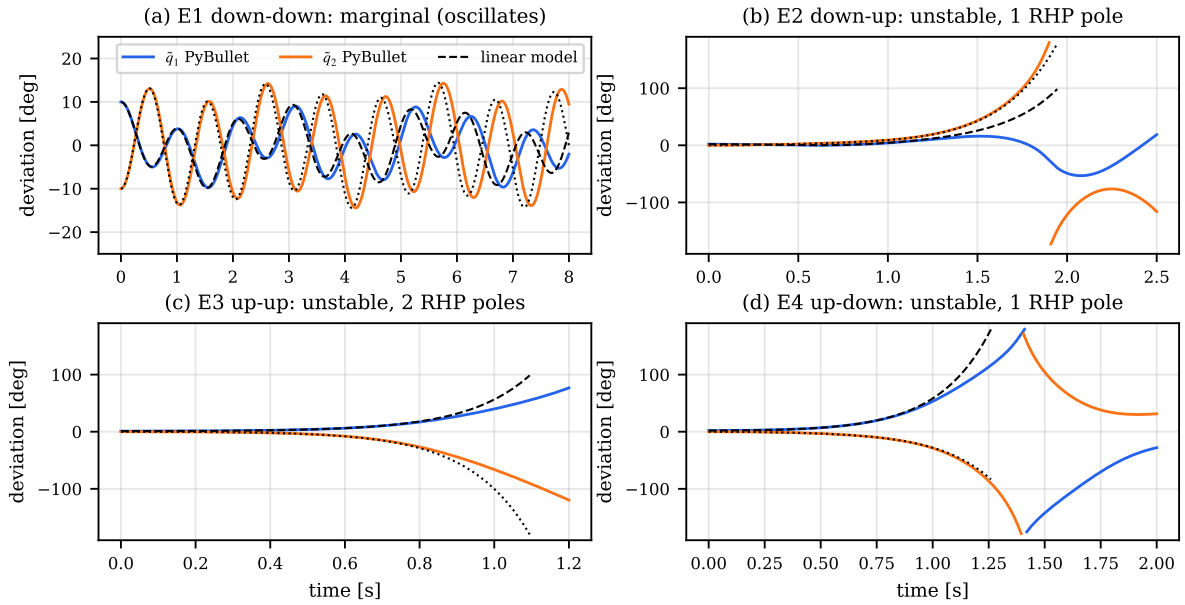


Figure 2: Free response ($\tau = 0$) near each equilibrium: PyBullet (colour) vs. linear model (black).

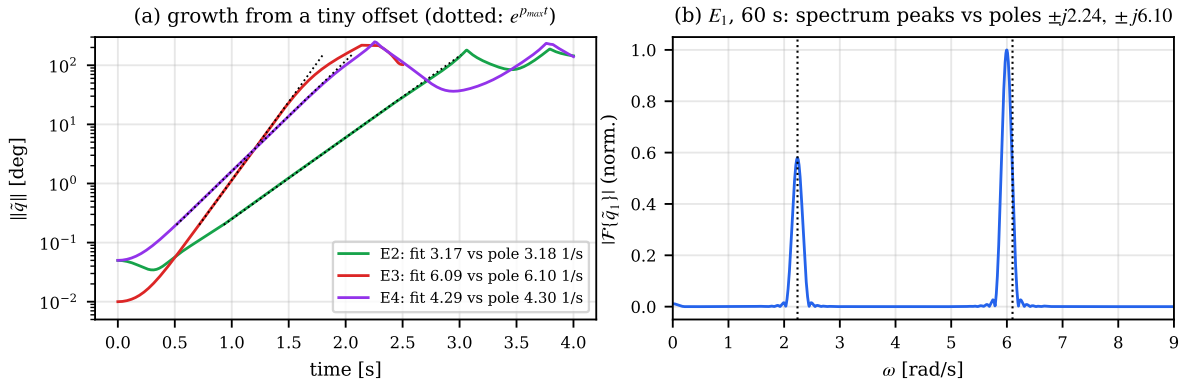


Figure 3: (a) Growth of a tiny offset on a log scale; dotted lines have the slope of the largest RHP pole. (b) Spectrum of $\tilde{q}_1$ near E_1; dotted lines are the poles.

Table 4: Linear predictions vs. PyBullet measurements

	Quantity	Predicted	Measured	Error
E_1	oscillation frequencies	2.240, 6.101 rad/s	2.239, 5.997 rad/s	within 1 bin
E_1	energy change over 60 s	0	1.2×10^{-3} J	integration error
E_2	growth rate	3.1775 s^{-1}	3.171 s^{-1}	0.22%
E_3	growth rate	6.1011 s^{-1}	6.088 s^{-1}	0.21%
E_4	growth rate	4.3010 s^{-1}	4.294 s^{-1}	0.15%

Observations:

- E_1 oscillates forever at its initial amplitude, with frequencies matching the poles.

- A tiny offset at E_2, E_3, E_4 grows exponentially at exactly the rate of the largest RHP pole. At E_3 a 0.01° tilt reaches 30° in 1.55 s.
- The linear model tracks the nonlinear one until about 20°, then over-predicts.

2.7 Can One Measured Angle Stabilize the Upright?

2.7.1 Static or PD Feedback

Measuring one angle $y = \tilde{q}_j$ and applying $\tau = -(K_p + K_d s)\, y$ with $G_j = (b_2 s^2 + b_0)/(s^4 + a_2 s^2 + a_0)$ gives the closed-loop polynomial

$$s^4 + (K_d b_2)\, s^3 + (a_2 + K_p b_2)\, s^2 + (K_d b_0)\, s + (a_0 + K_p b_0). \tag{11}$$

Hurwitz requires $K_d b_2 > 0$ *and* $K_d b_0 > 0$, i.e. b_2 and b_0 of the same sign. At E_3:

- measuring q_1: $b_2 = -3.0147$, $b_0 = +12.6926$, opposite signs;
- measuring q_2: $b_2 = +6.0332$, $b_0 = -31.7316$, opposite signs.

The signs are opposite because the zeros ±2.0519 and ±2.2934 are **real**: one of each pair lies in the right half-plane. **No K_p, K_d can stabilize E_3 from one angle.** A grid search confirms it: the smallest attainable largest real part is 0.96 (q_1) and 2.29 (q_2), and in PyBullet even these gains fall from 1° within 2 s.

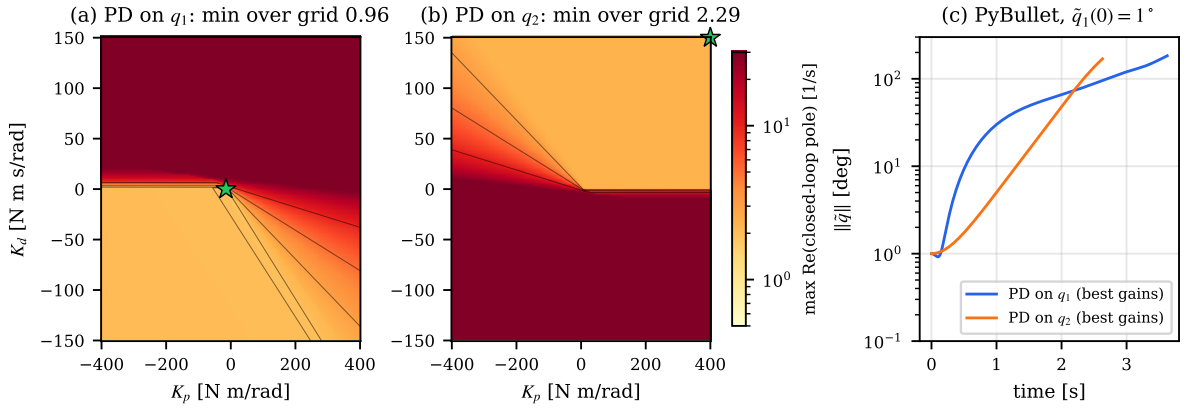


Figure 4: PD from one angle at E_3: (a, b) largest closed-loop real part over the gain plane, positive everywhere; (c) PyBullet with the best gains.

2.7.2 Dynamic (Observer-Based) Feedback

Each angle alone makes (A, C_j) observable, so an observer can reconstruct the full state and LQR can act on the estimate:

$$\dot{\hat{x}} = A\hat{x} + B_\ell \tau + L(y - C_j \hat{x}), \qquad \tau = -K\hat{x}, \qquad \lambda_{cl} = \lambda(A - B_\ell K) \cup \lambda(A - L C_j). \tag{12}$$

Two limits apply to *any* controller:

- **Parity interlacing [6]:** for q_1 the interval $(2.05, \infty)$ contains two real RHP poles, so a stable controller suffices. For q_2, $(2.29, \infty)$ contains one, so the controller itself must be unstable.
- **Sensitivity bound [3]:** $\|S\|_\infty \geq \prod_i |z + p_i|/|z - p_i|$, which is ≥ 45.9 for q_1 and ≥ 187 for q_2, because each zero sits just below the pole 2.24.

In PyBullet the observer-based loop holds E_3 only from 1.42° (measuring q_1) or 0.23° (measuring q_2), against 4.64° with full-state feedback.

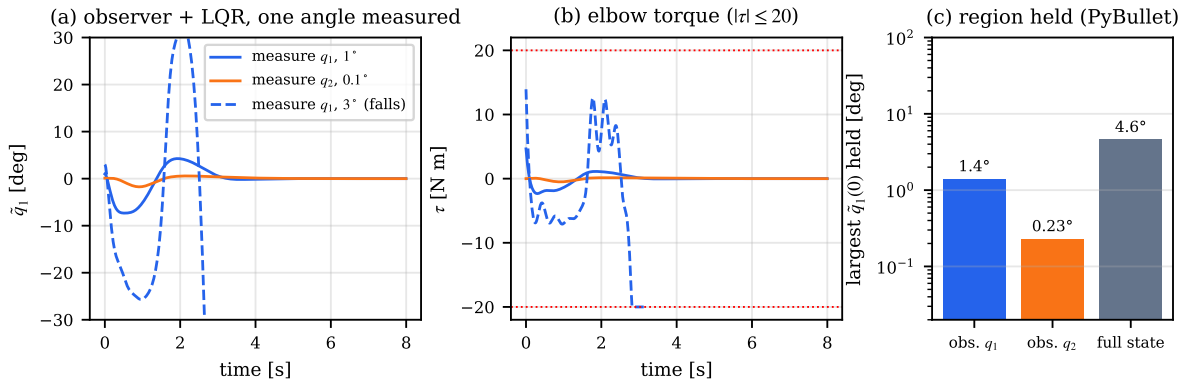


Figure 5: Observer + LQR at E_3 in PyBullet using one angle: (a) $\tilde{q}_1$; (b) torque; (c) largest initial offset held.

Answer: static or PD feedback from one joint angle **cannot** stabilize the upright, because of the real RHP zero. A dynamic controller **can**, but only for about a degree with $|\tau| \leq 20\,\mathrm{N\,m}$, so Phase 3 uses full-state feedback.

3 Simulations and Code

3.1 PyBullet Simulator

All simulations use the full nonlinear model in PyBullet with an interactive dashboard (Fig. 6). The user picks an equilibrium, an initial condition (slider, typed value or by dragging the arm) and a controller: open loop, PD on one angle (live K_p, K_d), or observer + LQR on one angle. The dashboard shows $\tilde{q}(t)$, $\tau(t)$ and the closed-loop poles live. File: `phase2/acrobot_phase2_sim.py`; instructions in `phase2/README.md`.

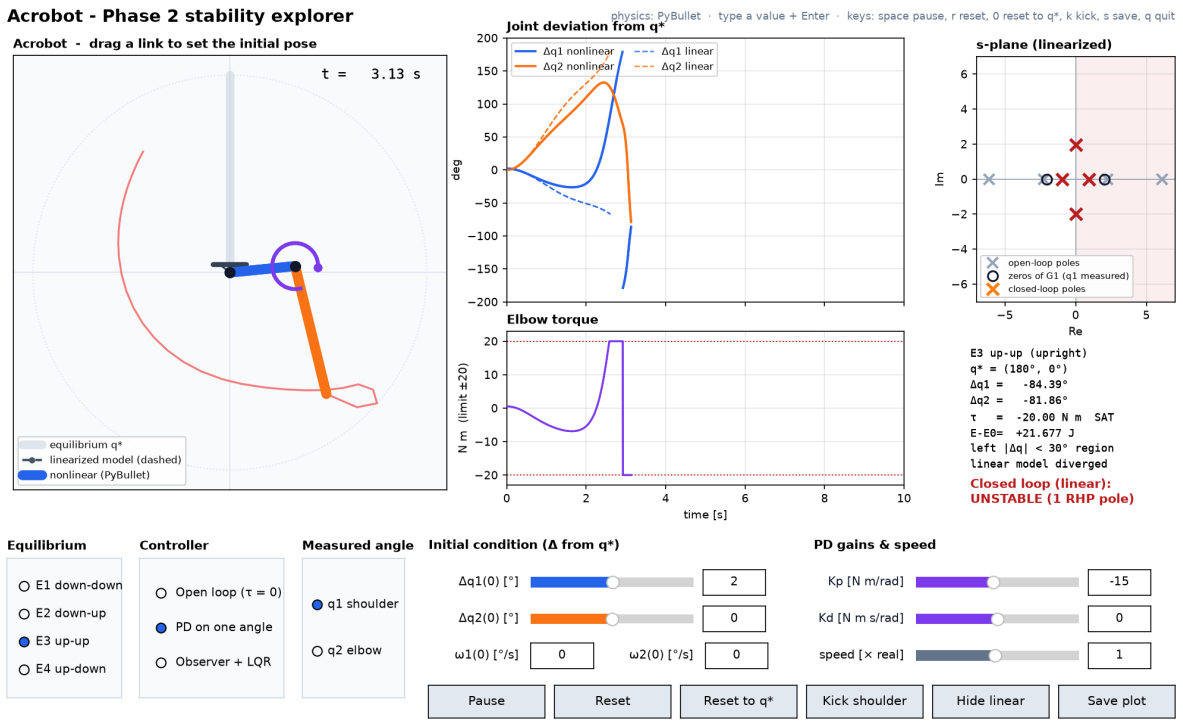


Figure 6: Phase 2 dashboard: PD on q_1 at the upright; a closed-loop pole stays in the right half-plane.

3.2 MATLAB Codes

It prints the poles, zeros, ranks and verdict for every equilibrium, measures the growth rates with `ode45`, and runs the single-angle PD search. Every number matches the tables above.

The files are listed below; the algorithms they implement follow as pseudo-code.

Table 5: MATLAB files

File	Purpose
`Acrobot_Nonlinear_Dynamics.m`	nonlinear model $\dot{x} = f(x, \tau)$ with the 20 N m limit (Phase 1)
`Acrobot_Linearize.m`	A, B_ℓ at an equilibrium (Phase 1)
`Acrobot_Phase2_Main.m`	poles, zeros, ranks, verdicts, growth rates, single-angle PD test

Algorithm 1 Nonlinear dynamics $\dot{x} = f(x, \tau)$ (`Acrobot_Nonlinear_Dynamics.m`) and linearization (`Acrobot_Linearize.m`)

function DYNAMICS($x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$, τ)
  $\tau \leftarrow \max(-20, \min(20, \tau))$ ▷ motor limit
  build $M(q)$, $C(q, \dot{q})$ and $\tau_g(q)$ from the given model
  solve $M\,\ddot{q} = \tau_g + B\tau - C\dot{q}$ for $\ddot{q}$ ▷ backslash, never M^{-1}
  return $[\dot{q};\ \ddot{q}]$
end function
function LINEARIZE(q^*)
  $K \leftarrow \partial\tau_g/\partial q$ at q^*; $P \leftarrow M(q^*)^{-1}K$; $r \leftarrow M(q^*)^{-1}B$
  return $A = \begin{bmatrix} 0 & I \\ P & 0 \end{bmatrix}$, $B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix}$
end function

Algorithm 2 Stability analysis and single-angle test (`Acrobot_Phase2_Main.m`)

for each equilibrium $E_k \in \{E_1, E_2, E_3, E_4\}$ **do**
  $(A, B_\ell) \leftarrow$ LINEARIZE(q_k^*); poles $\leftarrow \mathrm{eig}(A)$
  $G_1, G_2 \leftarrow$ transfer functions $\tau \to q_1$, $\tau \to q_2$; zeros $\leftarrow$ roots of their numerators
  check $\operatorname{rank}\mathcal{C} = \operatorname{rank}\mathcal{O}_1 = \operatorname{rank}\mathcal{O}_2 = 4$ ▷ minimal: poles of $G_j = \mathrm{eig}(A)$
  verdict $\leftarrow$ asymptotically / marginally stable / unstable from the signs of Re(poles)
end for
for $E_k \in \{E_2, E_3, E_4\}$ **do** ▷ nonlinear check
  simulate $\tau = 0$ from a 0.01° offset (`ode45`)
  fit the slope of $\ln\|\tilde{q}(t)\|$ and compare it with the largest RHP pole
end for
for measured angle $j \in \{1, 2\}$ at E_3 **do** ▷ PD from one angle
  write $G_j = (b_2 s^2 + b_0)/\Delta(s)$
  if $b_2\, b_0 < 0$ **then**
    PD is impossible (no Hurwitz K_p, K_d)
  end if
  grid-search (K_p, K_d): record the smallest $\max \mathrm{Re}\,\lambda(A - B_\ell F)$
end for

3.3 Animation

A 2K Manim animation of this phase: watch the video.

4 Tabulation

Table 6: Summary of the stability analysis

Equilibrium	RHP poles	Energy shape	Stability
E_1 down–down	0	minimum	marginally (Lyapunov) stable
E_2 down–up	1	saddle	unstable
E_3 up–up	2	maximum	unstable
E_4 up–down	1	saddle	unstable

Table 7: Single-angle feedback at the upright E_3 ($|\tau| \le 20\,\mathrm{N\,m}$)

Controller	Measuring q_1	Measuring q_2
PD (any K_p, K_d)	impossible (best $\max \mathrm{Re}\,\lambda = 0.96$)	impossible (best 2.29)
Observer + LQR	held up to 1.42°	held up to 0.23°
Stable controller possible?	yes	no (must be unstable)
Sensitivity bound $\|S\|_\infty \ge$	45.9	187

5 Conclusion

The acrobot has four zero-torque equilibria. Only the hanging pose E_1 is stable, and only in the Lyapunov sense: its poles lie on the imaginary axis and, with no friction, it oscillates forever. E_2 and E_4 have one RHP pole each (saddles of the potential energy) and the upright E_3 has two (a maximum), so all three are unstable. The transfer functions and the state matrix give the same verdicts because each realization is minimal, and PyBullet confirmed the predicted growth rates to within 0.25%.

A single measured angle cannot stabilize the upright with static or PD feedback: the real right-half-plane zeros of G_1 and G_2 make the required coefficient signs contradict each other. An observer-based dynamic controller can stabilize it, because each angle alone makes the system observable. With $|\tau| \le 20\,\mathrm{N\,m}$ it holds only about a degree, and via q_2 the controller would itself have to be unstable. This motivates full-state feedback in Phase 3.

6 Contributions

1. Report compilation – _____
2. Stability criteria and analysis (Sections 2.1–2.5) – _____
3. Simulation evidence (Section 2.6) – _____
4. Single-angle feedback (Section 2.7) – _____
5. PyBullet simulator and figures – _____
6. MATLAB codes – _____
7. Animation (Manim) – _____

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